B4.2 Ecological niches: data-based questions

NAME	DATE	TOTAL
		_____ / 6

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 Hawks and their prey

6 MARKS

In many birds of prey, females are larger than males. The mean body mass of males and females of four North American hawks was compared with the mean mass of the prey they caught.

SPECIES	MALE MASS / G	FEMALE MASS / G	MALE PREY MASS / G	FEMALE PREY MASS / G
SHARP-SHINNED HAWK	100	170	20	30
COOPER'S HAWK	280	440	60	120
NORTHERN GOSHAWK	900	1300	300	500
RED-TAILED HAWK	1000	1200	150	180

a COMPARE the body mass of males and females. [1]

b CALCULATE the ratio of prey mass to body mass for female Cooper's hawks. [1]

c DESCRIBE the relationship between the body mass of a hawk and the mass of its prey. [1]

d EXPLAIN how the difference in size between males and females could reduce competition between them. [2]

e SUGGEST one reason why the red-tailed hawk does not fit the pattern of the other three species. [1]